Ewa Miazga | Computer Vision Researcher

+48 662 700 920 | ewa.miazga02@gmail.com | GitHub | LinkedIn

About Me

Data Science Master's student at EPFL specializing in primitive-based algorithms for 3D Computer Vision, with a strong focus on efficient 3D reconstruction and Gaussian Splatting. Proven experience contributing to state-of-the-art research pipelines in collaboration with academic labs across Switzerland. Strong ability to conduct independent, publication-oriented research, from problem formulation to experimental validation. In my free time, I enjoy cycling, sailing and participating in hackathons.

Core Skills

- Programming Languages: Python, C++, Java, Scala, SQL, PL/SQL, C, CUDA
- Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Keras, SciPy, NetworkX, Qt (C++)
- Software & Tools: Git, Hugging Face, Bokeh, Seaborn
- Databases: MySQL
- Operating Systems: Strong Linux foundation
- Teamwork & Collaboration: Interdisciplinary research and hackathon experience
- Research & Experimentation: Computer Vision, gradient-based optimization, transformers capabilities, neural radiance fields
- Communication Presentations, mentoring, leadership, technical writing
- Languages: English - C1, French - A2, Polish - Native

Publications

Beyond Spherical Harmonics: Rethinking Appearance Models for Radiance Reconstruction — EGSR'26
Ewa Miazga, Jorge Condor, Piotr Didyk — *Bordeaux, France*

Work Experience

Computer Vision Lab Intern — Sep. 2025 – Present
Università della Svizzera italiana (USI) — *Lugano, Switzerland*

- Proposed a novel kernel - NASGabor for modeling complex structures on the sphere.
- Developed an appearance model within the 3D Gaussian Splatting pipeline, improving reflection fidelity and reducing memory usage by 66%.
- Derived and implemented analytical derivatives in CUDA, accelerating rendering.

AI Lab Student Intern — Sep. 2024 – Jul. 2025
École Polytechnique Fédérale de Lausanne (EPFL) — *Lausanne, Switzerland*

- Developed computer vision models for depth estimation and geometric object representation from limited real-world data.
- Designed graph-based object representations using Bézier curves, angle approximation, and network refinement.

Junior Software Engineer — Jul. 2022 – Feb. 2023
Redge Technologies — *Warsaw, Poland*

- Developed a C++/Qt application for OTT data-stream monitoring, improving system reliability and administrator user experience.
- Enhanced performance and ensured compliance with OS-level integration requirements.

Education

École Polytechnique Fédérale de Lausanne (EPFL) — Sep. 2024 – Jul. 2026
MSc Data Science (GPA: 5.36/6), Specialization: Computer Vision, Computer Graphics — *Lausanne, Switzerland*

- Conducted research on **3DGStream Object Reconstruction**, proposing a dynamic Gaussian densification and attention-aware optimization method to enhance novel object reconstruction quality in 3D scenes.
- Collaborated with the **EPFL–Yale LiGHT Lab** on **Context-dependent unlearning report** in large language models, evaluating AdaLora and Wise frameworks on Llama and Meditron.

École Polytechnique Fédérale de Lausanne (EPFL) — Sep. 2021 – Jun. 2024
BSc Computer Science (GPA: 5.22/6, Thesis: 6/6) — *Lausanne, Switzerland*

- Developed an **Interactive piano exoskeleton** prototype using 3D printing to assist pianists in targeted muscle memory training. Completing the project in a fast-paced, interdisciplinary 6-month sprint.
- Built a **Time-series forecasting model** for a Swiss EdTech startup to identify students at risk of academic failure. Managed big data pipelines, incorporating bias awareness to ensure robust and trustworthy tools.
- Researched **Continual learning techniques** (EWC, Familiarity AutoEncoders) to mitigate catastrophic forgetting in neural networks. Focusing on reducing the high computational costs inherent to training.